

Paper Manager pack — World Simulation / Cosmos WFM (Dec 7 · Accompanying Models)

Paper: World Simulation with Video Foundation Models for Physical AI
arXiv: <https://arxiv.org/abs/2511.00062> · [predict2.5](#) · [transfer2.5](#) · arXiv **2511.00062v2** (24 Feb 2026) · NVIDIA Open Model License
Authors: NVIDIA Cosmos / Physical AI (Arslan Ali ... Yuke Zhu; large author list — Sec. A)
PDF: /workspace/papers/world-simulation-2511.00062.pdf (= /workspace/world-simulation-quiet/paper.pdf) · **44 pp** · ~35 MB
Session: Dec 7 · due 9:30 AM · Type: **Accompanying Models (World Models, Reasoning, Planning)**
Role weights: **Archaeologist=3 (Nykolas Rekasius) LONGEST PRIMARY; Visionary=1 (Kanta Saito);** Stakeholder / Scientific Reviewer / Empiricist seats empty — **still all five produced;** Archaeologist heaviest
Kanta role: Visionary=1 seated — BEHAVIOR / closed-loop / photoreal synth hooks; dual-WM fork (generative video WFM vs latent LeWM/JEPA); **skip Discord/Drive**
Themes: generative video **WFM as physics/world simulator** (alt to latent WMs) · **Cosmos-Predict2.5** flow matching unified **T2W/I2W/V2W** · **Cosmos-Reason1** text · **Transfer2.5** Sim2Real/Real2Real (**3.5×** smaller than Transfer1-7B) · synth data / policy eval / closed-loop robotics+AV · **PARALLEL** to Ha/Dreamer / I-JEPA → V-JEPA → LeWM (latent) — **not** a JEPA child
Sources: Summarizer (/workspace/papers/world-simulation-2511.00062-summary.md) + Senior Research (/workspace/world-simulation-quiet/DELIVERABLE.md) + Ablation Bot (**GAP** — paper-extract Tabs. 10–20 / §6–7) + paper_notes.md / env_outline/
Sibling links: /workspace/leworldmodel-quiet/ · /workspace/vjepa21-quiet/ · /workspace/ijepa-quiet/ · /workspace/robogen-quiet/ (generative *curriculum* foil)

1. One-liner

Open Physical-AI generative-video world foundation models — **Cosmos-Predict2.5** (flow-matching DiT, unified **Text2World / Image2World / Video2World**, **Reason1** grounding, **2B / 14B**, ~**200M** curated clips + RL post-train) + **Cosmos-Transfer2.5** (control-net Sim2Real/Real2Real; **3.5×** smaller than Transfer1-7B yet higher fidelity + better long-horizon RNDS). Enables synthetic data, policy evaluation, and closed-loop robotics/AV simulation — **pixel-space world simulator** pole, **PARALLEL** to latent JEPA / Dreamer planners (LeWM / V-JEPA 2-AC), **not** their child.

Contrast poles (consistency triangle — extend prior packs; do not contradict):

Pole	What they do	Primary claim / use
DINOv2 / SigLIP	Discriminative JEA / dense towers	VLA vision towers; DINO-WM freeze
MAE / Dreamer / IRIS	Generative / reward-tied WMs	FT or RL imagination (latent/pixel)
I-JEPA → V-JEPA → 2 → 2.1 → 2-AC → LeWM	Predictive embedding / compact latent plan	Fast CEM / MPC in latent — PARALLEL foil
Cosmos-Predict1 → Predict2.5 + Transfer2.5	Generative video WFM as Physical AI simulator	Synth data, photoreal eval, AC video, Sim2Real — THIS

Do NOT invent / claim: total GPU-hours as a single headline (paper gives **4096 H100 MFU**, not aggregate GPU-h); course smoke = PAI **0.768** / Tab. 13 **24/30** / DreamGen VLA task SR; Cosmos as JEPA descendant;

2. Summary (exec overview + key numbers)

NVIDIA Cosmos (arXiv 2511.00062v2) ships open **World Foundation Models** for Physical AI. Core inventum: treat **large generative video** as a **physics / world simulator** — complementary to latent Dreamer/JEPA planners. **Predict2.5** replaces Predict1’s EDM diffusion with **flow matching**; unifies T2W/I2W/V2W via frame-replacement conditioning; swaps T5 for **Cosmos-Reason1** (Physical AI VLM → 1024-d multi-block); progressive pretrain 256 → 480 → 720p on **~200M** curated clips (from **35M hours** raw, **~4%** survival) + domain SFT soup + **VideoAlign RL** + optional **4-step** rCM distillation. **Transfer2.5** is a control-net (edge/blur/depth/seg) on Predict2.5-**2B** — **3.5×** smaller than Transfer1-**7B**, better adherence/quality, less long-horizon RNDS drop.

Why Accompanying Models / Dec 7 after LeWM: Students just saw **compact latent JEPA-WM** (LeWM ~15M, 0.98 s CEM). Now see the **opposite compute contract**: foundation-scale **pixel/video** imagination for synth data, photoreal policy eval, Sim2Real/Real2Real. Syllabus lesson = **the fork**, not flattening both into one “world model.”

Axis	Number / claim
Venue / code	arXiv 2511.00062v2 ; nvidia-cosmos/cosmos-predict2.5 + cosmos-transfer2.5 CLEAR
Scales	Predict 2B / 14B ; Transfer 2B (vs Transfer1 7B)
Frames	93 px / 24 latent @ 16 fps (~5.8 s); WAN2.1 VAE 4×8×8 + 1×2×2 patch
Data	~200M curated clips; robotics Tab. 2 (AgiBot/Bridge/DROID/GR00T/...)
PAI T2W Overall post (Tab. 10)	2B/14B 0.768 (pre 2B 0.751)
PAI I2W Overall post (Tab. 11)	2B/14B 0.810 (pre 2B 0.799)
Human pref 14B vs Wan2.1-14B (Fig. 7)	48.6% vs 31.8%
Distill (Tab. 7–8)	T2W 0.768 → 0.764 (4 steps); I2W 0.810 → 0.816
Transfer size / quality (Tab. 12)	3.5× smaller; Uniform Quality 9.31 vs Transfer1 9.24 ; Edge F1 0.49 vs 0.38
Robot Real2Real aug (Tab. 13)	Proposed 24/30 · Baseline 5/30 · Base 1/30
AV FVD / LET-AP (Tab. 14–15)	FVD 23.06 vs Predict1 63.69 ; LET-AP 0.394 vs 0.243
DreamGen Object GPT (Tab. 18)	Predict2.5-14B/gr00tdream-gr1 91.8 (video IF — not VLA SR)
Bridge AC (Tab. 19–20)	PSNR 24.95 / FVD 146 ; TimeEmb >> CrossAtten >> ChannelConcat
Infra (Tab. 9)	4096 H100 ; MFU 2B 36.49% / 14B 33.08% — course train = N

Must-cite tattoo: PAI T2W/I2W **0.768 / 0.810** · Transfer **3.5×** · robot **24/30 vs 5/30** · Bridge PSNR **24.95** · TimeEmb Tab. 20 · DreamGen **91.8** (IF only) · AV LET-AP **0.394** · **latent vs pixel fork** · **infer-only** course smokes.

Do NOT claim: course = paper PAI / 24/30 / 4096-H100 train; DreamGen IF = VLA task success; closed-loop = Dreamer-style actor trained purely in Cosmos imagination (paper: Transfer aug + AC video eval + long-horizon Transfer); Cosmos ∈ JEPA genealogy.

3. Keywords

Cosmos-Predict2.5; Cosmos-Transfer2.5; Cosmos-Reason1; world foundation model; flow matching; Text2World; Image2World; Video2World; Physical AI; synthetic data; policy evaluation; closed-loop simulation; Sim2Real; Real2Real; control-net; multi-view driving; action-conditioned video; DreamGen; Bridge; AgiBot; GR00T GR1; PAI-Bench; WAN2.1 VAE; latent vs generative WM; Accompanying Models; BEHAVIOR synth/eval (TENTATIVE)

4. Deep dive — World-model / planning + lineage triangle (PRIMARY SESSION CONTRIBUTION)

Callout: Scientific product of *this* PDF = open **generative-video Physical AI world simulator** (Predict + Transfer + Reason1 + apps). Session type = Accompanying Models because Dec 7 must teach **why latent planners and pixel WFM**s both exist — opposite compute contracts under the same “world model” phrase.

Taxonomy — CLEAR vs Ha/Dreamer, JEPA/LeWM, this paper

Family	State / objective	Control use	Course smoke	Foil role
Ha & Schmidhuber / Dreamer	Latent VAE+RNN / RSSM; imagine + reward	RL in imagination	Trainable small	Latent-paradigm parent (§7 cite)
I-JEPA → V-JEPA → 2.1 → 2-AC	Predict embeddings ; AC CEM	Real-robot latent plan (Franka)	Sibling packs	PARALLEL foil (§7 cites V-JEPA 2)
LeWM	Compact E2E JEPA + SIGReg; ~15M	Sim CEM 0.98 s (~48× vs DINO-WM)	Trainable 1 GPU	Compact latent foil
Classic physics sims (IsaacSim...)	Engineered dynamics	Train / closed-loop	—	Transfer photorealizes Sim2Real
Sora-class / Wan / Hunyuan / Cog	Generative video	Mostly content	—	PAI / DreamGen foils
Cosmos-Predict1 / Transfer1	Prior NVIDIA Physical AI WFM (EDM)	Synth / AV	—	Direct predecessor
THIS (Predict2.5 + Transfer2.5)	Flow DiT pixels + control-net	Synth data, AC eval, Real2Real, AV	Infer-only probes	Inventum

Consistency / lineage triangle (extend; do not fold into JEPA)

DINOv2 / SigLIP — discriminative JEA / dense towers — VLA vision towers / DINO-WM freeze
MAE / Dreamer / IRIS — generative / reward-tied WMs — FT or RL imagination
I-JEPA → V-JEPA → 2 → 2.1 → 2-AC — Meta predictive SSL → AC latent planning (Franka)
 ↳ PLDM → LeWM — compact E2E JEPA-WM from pixels (sim CEM)
Cosmos-Predict1 → Predict2.5 (+ Reason1) + Transfer2.5 — GENERATIVE VIDEO WFM as Physical AI simulator
 (PARALLEL pole: pixel/video imagination, synth data, closed-loop / policy eval)
RoboGen (side) — generative *task/scene/reward* curriculum in classic sim – not pixels

One-liner (fork): Latent WMs (Dreamer/JEPA/LeWM) compress to plan; Cosmos WFMs decompress to simulate and synthesize — same phrase, opposite compute contracts.

What is “world-model-ish” here — CLEAR (in-paper)

Piece	In-paper role	World-model reading
Flow DiT Predict2.5	Velocity net; unified T2W/I2W/V2W	Pixel/video dynamics ($P(\text{video} \mid \text{cond})$)
Reason1 text	Physical AI VLM → 1024-d cross-attn	Language-grounded world prompts
Transfer2.5	Edge/blur/depth/seg control-net	Controllable Sim2Real / Real2Real translation
Action-cond (Tab. 19–20)	TimeEmb best; Bridge AR chunks	Action-conditioned video for policy evaluation
Real-robot Tab. 13	Transfer Real2Real → DP 24/30	Synth observation factory (offline aug)
AV multiview Tab. 14–15	Map-conditioned Transfer; FVD / LET-AP	Closed-loop AV scenario simulation
DreamGen Tab. 18	Text → robot video IF 91.8 Object GPT	VLA demo synth pipeline (IF ≠ task SR)
Long-horizon RNDS Fig. 10	Transfer2.5 less drop vs Transfer1	Stability of chunked pixel imagination

What is not overclaimable

- Full **online pixel CEM / Dreamer-in-Cosmos** agent SR on Franka/BEHAVIOR — **not tabulated**; Transfer robot result is **offline synth-data aug**; AC model = **video prediction / eval** quality.
- DreamGen Tab. 18 = **video instruction-following** (VLM judges), **not** hardware VLA success.
- Equivalence to Newton-solver physics accuracy — qualitative coherence; Domain/VideoPhy still needed.
- Course students train 200M / 4096 H100 — **N**; smoke = frozen **2B** infer + stubs.
- π_0 / OpenVLA citations — **absent** (GR00T / DreamGen / Diffusion Policy **are** cited).
- Collapsing Cosmos into JEPa genealogy — **FALSE**; paper §7 explicitly adopts generative-video pole **against** latent WMs.

Architecture checklist (Empiricist / Archaeologist pin)

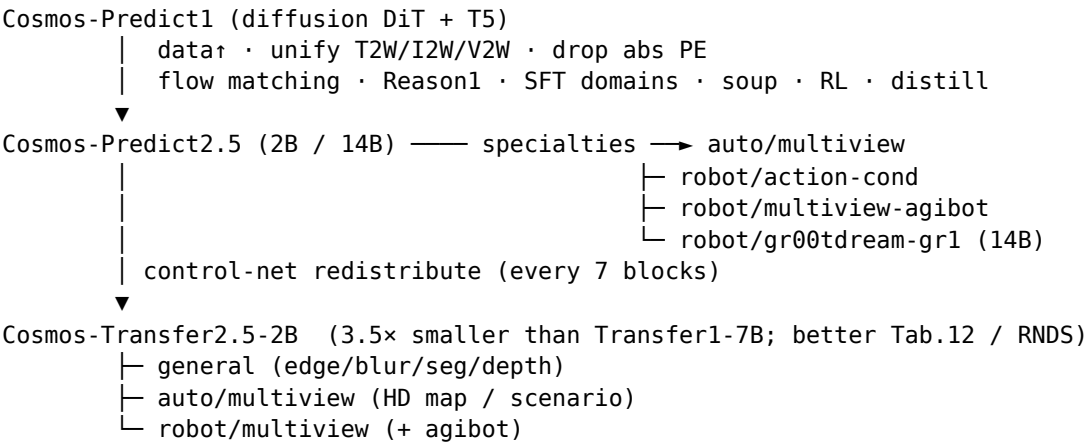
Piece	Spec (paper)
Paradigm	Pixel-space generative video WFM
Dynamics	Flow matching ($v_t = \text{varepsilon} \cdot x$); logit-normal + β shift 1 → 5; 5% high-noise oversample
Backbone	DiT velocity; 3D RoPE only (abs PE removed vs Predict1)
Scales	2B : 32L / d=2048 / 16H; 14B : 36L / d=5120 / 40H (Tab. 3)
VAE	WAN2.1 causal; 4×8×8 ; patch 1×2×2 ; 93 frames @16 fps
Text	Cosmos-Reason1 multi-block → 1024-d (replaces T5)
Modes	Unified T2W / I2W / V2W via frame-replacement + mask tokens
Post-train	Domain SFT (Tab. 5) → soup merge → VideoAlign RL → optional rCM 4-step distill
Transfer	4 control branches; blocks every 7 main blocks (vs Transfer1 prefix stack)
Action inject	MLP → time embeddings best (Tab. 20)
Infra	Tab. 9 4096 H100 MFU ~33–36% — course train N

World-model exam bite: "Cosmos-Predict2.5 = open Physical-AI **generative-video world simulator** (flow DiT + Reason1; T2W/I2W/V2W; PAI **0.768/0.810**); Transfer2.5 = **3.5×** smaller control-net with robot **24/30** Real2Real aug; Bridge AC **24.95** PSNR; **PARALLEL** to latent Dreamer/JEPa/LeWM — plan-fast vs synth-rich; course = **infer-only** probes, never 4096-H100 train."

5. Deep dive — Cosmos family evolution + ablation honesty (Archaeologist spine methods)

Callout: Ablation Bot extract **AUTHORITATIVE** at /workspace/papers/world-simulation-2511.00062-ablations.md. P0 #1 = **Tab. 20** action inject (only labeled ablation). Most “ablations” = system leaps (Predict1 → 2.5, Transfer1 → 2.5, Wan foils, Real2Real vs image aug). Empiricist must **not** pretend RC1 reproduces Tab. 10.

Family evolution diagram



Text stack: T5 → Cosmos-Reason1 (Physical AI VLM; multi-block 1024-d)

Ablation priority (paper-extract — Bot GAP)

Rank	Factor	Locus	Course smoke
1	Frozen Predict2.5-2B short-clip probe	Tabs. 10–11; release	Y — RC1 PRIMARY
2	Unified modes T2W/I2W/V2W same ckpt	§3.2; Tab. 1	Partial — RC2
3	Action inject TimeEmb / CrossAtten / ChannelConcat	Tab. 20	Partial — RC3 or cite
4	Closed-loop stub (policy ↔ WM)	§6.2 / §6.6	Partial — RC4
5	Transfer Edge vs Blur infer	Tab. 12	Partial if ckpt
6–11	Pre/post PAI; Transfer1 bake-off; Tab. 13; AV; DreamGen; distill	Tabs. 7–15, 18	N / cite
12	vs latent foils Dreamer/JEPA/LeWM	§7	Conceptual + tiny foil stub
13	Full Progressive / 200M / RL / 4096 H100	Tabs. 4–6, 9	N forever
GAP	Ablation Bot file	—	Fold when present

Taxonomy / recipe exam bite: "Predict1 → 2.5 is multi-factor (data+arch+Reason1+RL) — not a single course ablate; Tab. 20 TimeEmb **24.95** > Cross **24.41** > Concat **23.11** is the clean factorial; Transfer **3.5×** smaller yet better confounds base upgrade + block placement; course = frozen **2B** infer, never paper-scale train."

6. Role-ready sections (all five — Archaeologist RICHEST; Visionary=1 Kanta)

6.1 Archaeologist — WEIGHT 3 LONGEST PRIMARY (Nykolas Rekasius)

Seat charge: Deepen **history**, **foil framing**, **Cosmos family evolution**, **CLEAR/TENTATIVE discipline**. Place generative-video WFM pole carefully vs Ha/Dreamer and I-JEPA → V-JEPA → LeWM — **PARALLEL, not child**. Longest talk block (~**16–18** slides). Empiricist empty → invest in lineage richness; do **not** overclaim smokes. Ablation Bot **GAP** → paper-extract honesty.

Priors — CLEAR (cited / direct ancestry)

Prior	Role relative to Cosmos 2.5	Label
Cosmos-Predict1 / Transfer1 (NVIDIA 2025)	Immediate WFM predecessor (EDM DiT; T5; Transfer1-7B)	CLEAR
Flow matching Lipman'22; Esser SD3-style shifts	Training objective	CLEAR
EDM Karras'22	Predict1 foil parameterization	CLEAR
DiT + WAN2.1 VAE	Backbone + tokenizer	CLEAR
Cosmos-Reason1 (NVIDIA 2025)	Physical AI VLM text encoder (replaces T5)	CLEAR
Sora / Kling / Veo / MovieGen / Seedance	Closed Sora-class WFMs	CLEAR related
Wan / Hunyuan / CogVideoX / LTX	Open video gen foils (PAI / DreamGen)	CLEAR
Ha & Schmidhuber 2018; Dreamer (Hafner)	Latent WM paradigm foil	CLEAR (§7)
V-JEPA 2 Assran'25	Cited latent predictive WM example	CLEAR cited as foil — not ancestor
ControlNet-style / Transfer1	Transfer2.5 parent design	CLEAR
Diffusion Policy Chi; DreamGen Jang'25	Policy / synth-data consumers	CLEAR
Bridge / DROID / AgiBot / GR00T datasets	Robotics data priors	CLEAR
IsaacSim / classic physics engines	Sim2Real photorealization partner	CLEAR (app context)

Priors — TENTATIVE / design rhyme (do not overclaim)

- Treating Cosmos as “next I-JEPA / V-JEPA child”: **FALSE** — generate pixels ≠ predict embeddings.
- Equating Transfer Real2Real aug with V-JEPA 2-AC zero-shot Franka CEM: **different mechanism** (data aug vs latent plan).
- π_0 / OpenVLA as paper-cited consumers: **not cited** (rhyme ≠ citation).
- Pixel-space CEM with Predict2.5 as dynamics: research fork, compute-hostile vs LeWM/2-AC.

This paper — CLEAR inventum

1. Unified **flow-matching** Predict2.5 (**2B/14B**) Text/Image/Video2World with **Reason1** grounding.
2. Physical-AI-heavy ~**200M**-clip curation (35M h → ~4% keep) + domain SFT soup + VideoAlign **RL** + optional 4-step distill.
3. Transfer2.5 at **2B** beating Transfer1-**7B** on adherence/quality/long-horizon RNDS (**3.5×** smaller).
4. Application suite: robot Real2Real policy aug (**24/30**), AV multiview world-scenario maps, camera-controlled robot views, DreamGen GR1, Bridge AC policy-eval WM.
5. Open release under NVIDIA Open Model License (cosmos-predict2.5 + cosmos-transfer2.5).

Subsequent — CLEAR

Item	Note	Label
github.com/nvidia-cosmos/cosmos-predict2.5	Code + checkpoints	CLEAR
github.com/nvidia-cosmos/cosmos-transfer2.5	Transfer stack	CLEAR
Specialty ckpts (Tab. 1)	auto/mv, action-cond, agibot, gr00tdream-gr1	CLEAR
Prior-pack JEPA chain + LeWM	Remain valid parallel tracks	CLEAR unchanged

Subsequent / parallel — TENTATIVE

Item	Note	Label
BEHAVIOR-1K boards driven by Cosmos-aug VLAs	Plausible, not measured here	TENTATIVE
Pixel CEM planning with Predict2.5 dynamics	Research fork; compute-hostile	TENTATIVE
Distilled 4-step as real-time closed-loop renderer	PAI near-parity; robot Hz not reported	TENTATIVE
Dual-WM stack (LeWM plan + Cosmos synth/eval)	Visionary composition	TENTATIVE — not in-paper as stacked system
RoboGen × Cosmos	Task factory × pixel factory	TENTATIVE composition

Genealogy diagram (CLEAR framing — Dec 7 teaching map)

Pole A – Discriminative towers: SigLIP/DINO → VLA perception
Pole B – Predictive embeddings: I/V-JEPA → LeWM / 2-AC latent plan
[packs: ijepa-quiet / vjepa21-quiet / leworldmodel-quiet]
Pole C – Generative video WFM: Sora-class → Cosmos-Predict1 → ****Predict2.5+Transfer2.5 (THIS)****
as simulator / synth / eval
Side – RoboGen: generative curriculum in classic sim
[pack: robogen-quiet]

Syllabus after LeWM: teach **why B and C both exist** — B for **fast** imagination/planning; C for **rich** observation synthesis and visual domain randomization. **Do not collapse C into B.**

Archaeologist depth — paradigm foil card (\$7)

Axis	Latent WM (Dreamer / JEPA / LeWM)	Generative video WFM (Cosmos)
State	Compact embedding / RSSM / CLS	VAE latent decoded to pixels
Objective	Predict next repr (+ anti-collapse)	Flow/diffusion match video distribution
Control use	Fast MPC/CEM in latent (LeWM 0.98s)	Action-cond video; policy eval / synth demos
Synth data	Weak for appearance diversity	Strength — Transfer Real2Real / DreamGen
Course smoke	Trainable on 1 GPU (sibling packs)	Infer-only probes (this pack)
Failure mode	Collapse / poor visuals	Drift / hallucination / huge cost / imperfect physics

Reviewer notes (Archaeologist lens)

- 1. **Paradigm naming trap:** Always tag **latent-plan** vs **pixel-sim**.

- 2. **Multi-factor leaps:** Predict2.5 gains $\neq$ single ablate; RC1 $\neq$ Tab. 10.
- 3. **Human vs auto:** 2B $\approx$ 14B on PAI Overall; human pref still jumps with size (33% $\rightarrow$ 48.6% vs Wan2.1).
- 4. **Transfer “3.5 $\times$ smaller yet better”** confounds Predict2.5 base upgrade + Transfer redesign.
- 5. **Physics humility:** Photoreal $\neq$ guaranteed physics.
- 6. **Closed-loop:** Transfer long-horizon + robot BC aug + AC AR — **not** full Dreamer-in-Cosmos SR curves.
- 7. **Open weights $\neq$ course trainability** — 4096 H100 MFU is a **blocker**.
- 8. **Ablation Bot GAP** weakens Empiricist authority vs LeWM/V-JEPA packs — fold when present.

Archaeologist one-liner: *From Ha’s dreams $\rightarrow$ Dreamer’s latent imagination $\rightarrow$ JEPA’s representation prediction $\rightarrow$ LeWM’s stable compact planner $\rightarrow$ Cosmos WFM’s photoreal Physical AI simulator: Dec 7 should teach the fork, not flatten it.*

Archaeologist exam bite: "Cosmos-Predict2.5 = open Physical-AI **generative-video WFM** pole — successor to Predict1, **PARALLEL** to JEPA/LeWM (not child); inventum = flow+Reason1+unified modes+Transfer2.5@2B + apps (**24/30**, AV, DreamGen, Bridge AC); always tag latent-plan vs pixel-sim; Ablation Bot GAP $\rightarrow$ Tab. 20 + infer probes; never claim 4096-H100 course train."

6.2 Visionary — WEIGHT 1 (Kanta Saito) — BEHAVIOR’26 + closed-loop / photoreal synth

Seat charge: Decision-oriented BEHAVIOR / motion-planning / long-horizon / closed-loop hooks. Empiricist empty $\rightarrow$ **do not demand Tab. 13 replication**. Dual-WM stack framing. **No Discord/Drive checklist.**

BEHAVIOR connection (honest)

- Winning BEHAVIOR stacks lean $\pi_{0.5}$ / **FM action experts + strong VLM towers** — Cosmos enters as **observation factory** and **eval harness**, not as the motor policy.
- Paper **CLEAR** path: Transfer2.5 Real2Real visual randomization (colors, lighting, distractors, backgrounds) — maps to BEHAVIOR visual/domain shift. In-paper evidence: **24/30** vs **5/30** classic augs on one bimanual Kinova task (n=3 $\times$ 10 scenarios).
- DreamGen-style text $\rightarrow$ robot video $\rightarrow$ pseudo-action (IDM) for VLA data: aligned with scarce BEHAVIOR demos — but Tab. 18 is **video IF**, not BEHAVIOR SR.
- Multi-view head $\rightarrow$ gripper synthesis (AgiBot): egocentric household robots with wrist cams / occlusions.
- **Do not invent** BEHAVIOR-1K scores for Cosmos.

When to use generative video WFM vs latent LeWM/JEPA

Need	Prefer	Defer
Novel visual generalization for BC/VLA	Transfer2.5 Real2Real (+ real teleop mix)	Training Cosmos from scratch
Fast stage-level planning	LeWM / V-JEPA CEM	DiT video inside MPC
Task/scene diversity in classic sim	RoboGen -style propose $\rightarrow$ verify	Pixel WFM as task author
Photoreal eval videos / human prefs	Predict2.5 infer (2B first)	14B until needed
AV multiview what-if	Transfer/Predict auto/multiview	Course smoke
Course proof	Sibling latent smokes + this pack RC1/RC2/RC4	PAI absolute / Tab. 13

Motion-planning / long-horizon / closed-loop hooks

1. **Generative eval harness:** rollout policies into Predict2.5/Transfer worlds; score visual success / VideoAlign / detectors before real BEHAVIOR trials.
2. **Long-horizon Transfer:** RNDs-stable chunked video as backdrop for multi-step household scripts (still need a planner/policy).
3. **AC Predict2.5:** action-conditioned imagination for short-horizon **what-if** (Bridge metrics); complementary to V-JEPA 2-AC latent CEM — richer but slower.
4. **Hybrid (TENTATIVE research, not in-paper):** $\pi_{0.5}$ /FM executes motor chunks; **LeWM / V-JEPA 2-AC** does fast latent subgoal search; **Cosmos** supplies (a) synth demos, (b) photoreal eval, (c) optional slow generative imagination for rare events.
5. Camera-controllable multiview: fill occlusions for long-horizon rearrange.

Synthetic data for $\pi_{0.5}$ / FM stacks (careful)

Paper cites DreamGen / GR00T GR1 / Diffusion Policy — **not** π_0 by name. Legitimate Visionary rhyme: same **synth observation** → **policy** pattern generalizes to FM VLAs **if** IDM/action labels align — label **TENTATIVE transfer of idea**, not paper claim.

Visionary bets (weight 1)

1. **Dual-WM stack:** latent planner **plus** generative synth/eval — don't pick one religion.
2. **Treat Cosmos as data/eval appliance** in BEHAVIOR; treat LeWM/JEPA as **trainable** curriculum.
3. **Mix card:** real teleop : Transfer-aug : RoboGen-sim — Empiricist leave-one-out later (other packs).
4. **Non-goals:** course students won't reproduce 4096-H100 pretrain; success = correct **tool choice** + honest probes.

Follow-up research agenda (Kanta)

1. BEHAVIOR visual OOD suite with Transfer2.5 prompts → measure VLA SR lift.
2. Closed-loop: policy ↔ Transfer world-scenario / robot multiview at fixed Hz with distilled 4-step model.
3. Hybrid generative-eval + latent-plan tournament (Cosmos judge, LeWM/2-AC propose).
4. Physics-stress prompts as BEHAVIOR safety scenarios.
5. Compare AC Cosmos rollouts vs V-JEPA 2-AC CEM on shared Bridge/DROID episodes (video metrics vs control success).

New apps (beyond Discord/Drive)

- Photoreal Sim2Real overlay for Isaac/BEHAVIOR scenes before robot time.
- Multi-cam synthetic wrists for single-head demo expansion.
- Policy regression testing in generated kitchens (visual unit tests).
- AV-style world-scenario maps adapted to indoor semantic floorplans.

Composition sketch (not a job):

teleop/OXE demos → optional **RoboGen** task expand → **Transfer2.5** appearance expand → train **DP/ $\pi_{0.5}$** → **Predict action-cond** for imagination eval → real deploy. Latent **LeWM/JEPA** optional for high-frequency planning where pixels are too slow.

Visionary one-liner: *For 2026 BEHAVIOR, use Cosmos WFM when you need photoreal synthetic eyes; use LeWM/JEPA when you need fast trainable imagination — Dec 7's lesson is the fork, with Archaeologist richness and Empiricist infer-only humility.*

Visionary exam bite: "Cosmos = BEHAVIOR's **generative synth/eval layer** beside $\pi_{0.5}$ motors and JEPA latent planners; cite Tab. 13 **24/30** as Real2Real moral (one task); DreamGen **91.8** = video IF not VLA SR; hybrid dual-WM = **TENTATIVE proposal; no Discord/Drive.**"

(No Discord / Drive checklist content.)

6.3 Empiricist — produced despite empty seat (RCs from DELIVERABLE)

Seat empty this session. Produce **course smoke cards** anyway for Paper Manager continuity. **Tiny proxies only:** frozen API/ckpt probe, short clip generation, closed-loop stub. **Not** full WFM train. Bar = honesty vs LeWM pack (which *can* train tiny WMs) — Cosmos cannot. Ablation Bot **GAP**.

Claim under measurement

Frozen Predict2.5-**2B** generates short clips without crash; unified modes run; optional closed-loop stub finishes with finite loss — **directional, not PAI 0.768** / Tab. 13 **24/30**.

Hypotheses (falsifiable at smoke scale)

- 1. **H-gate (#1):** Released Predict2.5-2B generates ≥ 1 short clip ($\leq 93f$ or truncated) without OOM/crash on **1×24–80 GB** GPU; optional quality proxy non-degenerate.
- 2. **H-modes (#2):** Same ckpt accepts T2W, I2W, V2W; I2W/V2W preserve first-frame identity better than T2W (qualitative).
- 3. **H-inject (#3):** If action-cond code+ckpt: TimeEmbedding $\geq$ CrossAtten $\geq$ ChannelConcat on Bridge-lite **directional** (Tab. 20). Else: **cite-only**.
- 4. **H-loop (#4):** Closed-loop stub finishes *K* steps with finite loss; open-loop MSE rises with *H*. **Not** Tab. 13 SR.
- 5. **H-transfer (#5 Partial):** Edge/Blur Transfer infer respects control more than unconditional Predict.
- 6. **H-foil (#12):** On tiny control proxy, LeWM/JEPA latent plan is **orders of magnitude** cheaper than DiT video roll — generative WFM reserved for synth/eval, not inner MPC (Archaeologist framing; **not** a paper claim).

Falsifiers: ckpt unloadable; modes missing; TimeEmb worse than ChannelConcat on matched Bridge-lite; stub NaN in $<K$ steps; Transfer ignores edge; claiming train ablate / PAI absolute from smoke.

Prioritized matrix (paper-extract + DELIVERABLE)

#	Experiment	Paper hook	Smoke	Pri
RC1	Frozen Predict2.5-2B shortclip	Tabs. 10–11	Y — PRIMARY	P0
RC2	Modes T2W/I2W/V2W	§3.2; Tab. 1	Partial	P0
RC4	Closed-loop stub	§6.2 / §6.6	Partial	P0
RC3	Action inject Tab. 20	Tab. 20	Partial / cite	P1
RC5	Transfer Edge/Blur infer	Tab. 12	Partial	P1
—	Pre/post PAI train; Transfer1 bake-off; Tab. 13; AV; DreamGen; 14B; 4096-H100	—	N	Skip

Min package: RC0 → RC1 → RC2 → RC4 → optional RC3/RC5 → cite #6-#11 → conceptual foil #12.

Exact reproduce stack (EDIT)

```
# Official
# git clone https://github.com/nvidia-cosmos/cosmos-predict2.5
# optional: cosmos-transfer2.5
# Pin commit → logs/pins.txt ; NVIDIA Open Model License

# Course scout env (quiet outlines – preferred)
export COSMOS_WORK_ROOT=$SCRATCH/cosmos-wfm-smoke
cd /workspace/world-simulation-quiet
bash env_outline/setup_mamba.sh
# mamba activate cosmos-wfm-scout
mkdir -p logs runs checkpoints
# EDIT: download Predict2.5-2B (+ Reason1 / VAE) per upstream README
# Outlines only – do NOT sbatch from quiet box
# NEVER claim paper PAI 0.768 / Tab.13 24/30 / DreamGen VLA SR from smoke
# NEVER launch full WFM / Transfer / 14B / 4096-H100 train
```

Run cards (from DELIVERABLE)

RC0 — env_smoke (CPU/GPU gate)

Field	Spec
Goal	Clone predict2.5 (pin); mamba env; import + CUDA; list ckpts
Pass	Env builds; logs/pins.txt; GPU seen; no train
Budget	<0.5 h
Script	env_outline/slurm_env_smoke.sh

RC1 — ckpt_probe_shortclip (#1 — PRIMARY)

Field	Spec
Paper	Tabs. 10–11 PAI; release 2B post-trained
Goal	Load Predict2.5-2B ; generate 1–4 short clips (≤16–93f; downscale 480p/256p if needed)
Metrics	Crash-free; wall-time/clip; VRAM peak; optional CLIP/aesthetic; save mp4 + prompt json
Pass	≥1 clip written; no NaN; VRAM documented — not required: PAI 0.768
Budget	~0.5–4 GPU-h ; VRAM 24–80 GB ; 14B = N
Script	env_outline/slurm_ckpt_probe_shortclip.sh

RC2 — mode_t2w_i2w_v2w (#2 Partial)

Field	Spec
Paper	§3.2 unified modes
Goal	Same seed/steps: T2W vs I2W (1 frame) vs V2W (2–8 cond frames)
Pass	All three paths run; I/V identity >> T2W to cond frame
Budget	~1–6 GPU-h
Script	env_outline/slurm_mode_t2w_i2w_v2w.sh

RC3 — action_inject_ablation (#3 Partial / cite)

Field	Spec
Paper	Tab. 20 Bridge PSNR/SSIM/LatentL2/FVD
Goal	Reproduce TimeEmb > CrossAtten > ChannelConcat or document cite-only
Pass	Directional ranking or explicit “cite-only — no inject swap in release”
Budget	Cite 0 ; lite FT ~2-12 GPU-h hard cap
Script	env_outline/slurm_action_inject_ablation.sh

RC4 — closed_loop_stub (#4 Partial)

Field	Spec
Paper	§6.2 policy+Transfer; §6.6 action-cond AR
Goal	obs → (optional policy) → action → WM predict chunk → next obs for $K=5-20$
Env	Bridge-lite replay or toy arm; no real robot
Pass	Loop completes; log foil cost note vs LeWM/JEPA
Budget	~1-8 GPU-h
Script	env_outline/slurm_closed_loop_stub.sh
Cross-link	LeWM RC3/RC5 planning speed; V-JEPA robot CEM

RC5 — transfer_infer_edge_blur (#5 Partial, optional)

Field	Spec
Paper	Tab. 12; Fig. 9-10
Goal	Infer Transfer2.5 Edge vs Blur on 1-5 clips; log adherence proxies
Budget	~1-6 GPU-h
Script	env_outline/slurm_transfer_infer_edge_blur.sh

Numbers to memorize (exam)

Fact	Value
PAI T2W / I2W post	0.768 / 0.810
Human 14B vs Wan2.1-14B	48.6% vs 31.8%
Transfer shrink	3.5× (2B vs 7B)
Robot Tab. 13	24/30 vs 5/30 vs 1/30
Bridge AC PSNR / FVD	24.95 / 146 vs Predict1 21.14 / 190
Action inject Tab. 20	TimeEmb 24.95 > Cross 24.41 > Concat 23.11
DreamGen Object GPT	91.8 (IF only)
AV LET-AP / FVD	0.394 vs 0.243 · FVD 23 vs 63
Distill	4 steps; T2W 0.768 – 0.764
MFU Tab. 9	4096 H100 ; 2B 36.49% / 14B 33.08%

Fact	Value
Scope	Infer-only course · no full train · Ablation Bot GAP

Failure modes cheat-sheet

Symptom	Likely cause	Next action
OOM 2B	720p × 93f × full steps	Drop res/frames/steps; enable offload
Reason1 missing	Aux ckpt not fetched	Fetch text encoder; pin path
Modes fail	Wrong cond packing	Match frame-replacement API
Tab. 20 unreproducible	Inject head not exposed	Mark cite-only
Loop NaN	AR latent drift	Teacher-force first frames; shorten <i>H</i>
Claiming train ablate	Seat empty + huge model	Relabel N ; stick to infer smokes
Claimed PAI 0.768 / 24/30 from smoke	Overclaim	Report directional / crash-free only

What not to invent / claim

- **Do not** claim course smoke = PAI **0.768** / Tab. 13 **24/30** / DreamGen VLA SR.
- **Do not** invent aggregate GPU-hours beyond Tab. 9 MFU snapshot.
- **Do not** launch full Progressive / RL / 14B / 7-cam AV / real-robot from course queue.
- **Do not** present Cosmos as JEPa descendant or as drop-in CEM replacement for LeWM/2-AC.
- **Do not** launder DreamGen IF into hardware VLA success.

Empiricist exam bite: "Quote **0.768/0.810, 24/30, 24.95** PSNR, **3.5×** Transfer, Tab. 20 TimeEmb; Empiricist RC1+RC2+RC4 = frozen **2B** infer + stubs — never claim paper absolutes / 4096-H100 train / DreamGen=VLA SR; Ablation Bot **AUTHORITATIVE** (Tab.20 P0)."

6.4 Stakeholder — produced despite empty seat

Buy / why schedule Dec 7 after LeWM: - Completes the WM triangle: compact latent JEPa-WM (LeWM) → foundation-scale **generative video** WFM.

- Direct Physical AI product relevance (NVIDIA open Cosmos stack) — robotics + AV demos in one paper.
- Hard numbers: **24/30** robot, PAI vs Wan, Transfer **3.5×**, DreamGen GR1, Bridge AC.
- Open weights → reproducible homework (generate / transfer / AC Bridge infer).

Risks / costs / limits: - **44-page systems** paper — Archaeologist must curate; not a clean 8-page idea paper.

- Compute inaccessible for from-scratch pretrain (**4096 H100** MFU); students consume checkpoints.
- Easy category error: calling it “a JEPa” or “replaces V-JEPa 2-AC planning.”
- Some “closed-loop” language is capability framing; primary quantified robot result is **offline aug**, not online MPC.
- Author list / Cosmos brand can overshadow scientific claims — keep Empiricist numbers front.
- Empiricist seat empty — RCs are proxies; Visionary=1 must not demand Tab. 13 lab.

Decision ask: Required companion after LeWM / V-JEPa for “generative simulator counterweight”; pair slides: LeWM (latent, fast, sim) | V-JEPa 2-AC (latent, real Franka) | **Cosmos (pixel WFM, synth+eval)**. Optional Dreamer as historical latent RL imagination bridge.

Stakeholder exam bite: "Buy Cosmos as the **generative-simulator counterweight** to LeWM — high syllabus value; police the JEPA-category error; students infer checkpoints, never pretend 4096-H100 course train."

6.5 Scientific Reviewer — produced despite empty seat

Claims under review

- 1. Flow-matching unified Predict2.5 + Reason1 yields a usable Physical AI video WFM at 2B/14B.
- 2. Transfer2.5 at 2B beats Transfer1-7B on adherence/quality/long-horizon.
- 3. Applications: Real2Real policy aug, AV multiview, DreamGen, Bridge AC policy-eval.
- 4. Generative video WFM is a coherent **paradigm alternative** to latent WMs (§7).

Strengths

- 1. Coherent Physical AI WFM narrative with open release.
- 2. Method upgrades concrete: FM, unified conditioning, Reason1, domain soup, RL, distillation.
- 3. Broad eval: PAI-Bench, human prefs, Transfer metrics, robot hardware, AV detection, DreamGen, Bridge AC.
- 4. Honest related-work split of latent vs pixel WM paradigms (cites V-JEPA 2 correctly as other pole).
- 5. Tab. 20 clean within-family action-inject factorial.

Weaknesses / ask-for-revision

- 1. Systems-paper density; FM vs EDM without matched retrain table.
- 2. Robot n=3 trials × 10 scenarios (30) — suggestive but small; single apple-bowl task.
- 3. DreamGen table ≠ VLA success; risk of over-reading.
- 4. “Closed-loop simulation” more Transfer stability + AC rollouts than full agent-in-loop SR curves.
- 5. Heavy proprietary curation (3.1M driving clips) — data reproducibility limited even if models open.
- 6. Comparison fairness: Wan sizes differ; human votes help but don’t replace matched compute.
- 7. Ablation Bot GAP for course packs; multi-factor Predict1 → 2.5 leap.

Verdict: Strong **systems + applications** contribution for Physical AI video WFMs; scientifically place as **generative world simulator**, not as latent planner SOTA. Accept for Accompanying Models as **required counterweight** after LeWM; demand precise claims on synth-data vs online planning.

Reviewer exam bite: Demand Tabs. 10–13 + 19–20 + Fig. 7 before slogan; cite **24/30** not bare “robot works”; punish DreamGen-IF-as-VLA-SR and JEPA-as-parent; fence closed-loop language to Transfer/AC evidence.

7. Empiricist run pack (env / scripts / GPU — from DELIVERABLE)

Env setup

```
export COSMOS_WORK_ROOT=$SCRATCH/cosmos-wfm-smoke
cd /workspace/world-simulation-quiet
bash env_outline/setup_mamba.sh
# mamba activate cosmos-wfm-scout
mkdir -p logs runs checkpoints
# EDIT: clone nvidia-cosmos/cosmos-predict2.5 (+ optional transfer2.5)
# Pin commit → logs/pins.txt
# Download Predict2.5-2B (+ Reason1 / VAE) per upstream README
# Outlines only – do NOT sbatch from quiet box
```

File	Role
env_outline/conda_env.yaml	cosmos-wfm-scout sketch

File	Role
env_outline/setup_mamba.sh	Create/update env
env_outline/run_checklist.md	RC0 → RC5 order
env_outline/README.md	Cluster usage, Ablation Bot GAP, siblings
env_outline/slurm_env_smoke.sh	RC0
env_outline/slurm_ckpt_probe_shortclip.sh	#1 PRIMARY
env_outline/slurm_mode_t2w_i2w_v2w.sh	#2 Partial
env_outline/slurm_action_inject_ablation.sh	#3 Partial/cite
env_outline/slurm_closed_loop_stub.sh	#4 Partial
env_outline/slurm_transfer_infer_edge_blur.sh	#5 Partial

GPU / RAM / TIME budget honesty

Assumptions: **infer / tiny stub** on Predict2.5-**2B** (or Transfer2.5-2B). Paper-scale video WFM train = **N**.

Workload	GPU VRAM	Host RAM	Wall time (order)	Smoke
RC0 env	— / 1 GPU import	16–32 GB	<0.5 h	Y
RC1 shortclip 2B	24–80 GB	64–128 GB	0.5–4 h	Y PRIMARY
RC2 modes ×3	same	64 GB	1–6 h	Partial
RC3 cite	—	—	0	cite
RC3 lite FT (only if forced)	40–80 GB	128 GB	2–12 h hard cap	Partial
RC4 closed-loop stub	24–80 GB	64 GB	1–8 h	Partial
RC5 Transfer infer	24–80 GB	64 GB	1–6 h	Partial
Paper 2B/14B pretrain + SFT + RL	4096×H100 class (Tab. 9)	cluster	weeks / N	N
Tab. 13 real robot	lab	—	N	N
Latent foil LeWM smoke (sibling)	10–20 GB	32 GB	hours (other pack)	sibling

Course envelope: $\leq \sim 8\text{--}20$ GPU-h **infer** for RC1+RC2+RC4 (+ optional RC5). Document aggressive downscale — don't pretend 720p/93f on consumer GPU.

Ruthless N list: full Progressive Tab. 4, domain SFT 30k×256, RL, 14B any train, 7-cam AV, DreamGen 14B, Transfer1 bake-off train, real-robot 24/30, 4096-H100 pretrain.

Success criteria vs paper claims

Paper claim	Minimal course criterion	Not required
Predict2.5 ships usable WFM	RC1: shortclip OK on 2B	Match PAI Overall ~ 0.77
Unified T2W/I2W/V2W	RC2: three paths run	Identity metrics
Reason1 text grounding	Prompt follow qualitative	Human/VQA stub
Transfer 3.5× smaller + better	Cite Tab. 12 / Fig. 10	Infer Edge/Blur RC5
Robot synth aug helps	Cite Tab. 13 24/30	Lab replication
AV multiview quality	Cite Tab. 14–15	—

Paper claim	Minimal course criterion	Not required
Action-cond > Predict1	Cite Tab. 19; RC3 Tab. 20	Lite Bridge PSNR
DreamGen VLA synth	Cite Tab. 18 (IF)	Hardware VLA SR
Pixel WFM vs latent WM	Archaeologist \$7 + foil note	Bake-off
Distill retains quality	Cite Tab. 7 0.764≈0.768	—
Paper-scale train	Document N	Never course

Pass: RC0+RC1 outlines runnable when cluster+ckpt available; RC2+RC4 Partial; rich Archaeologist \$6.1; Visionary \$6.2; no jobs launched in quiet.

Fail / overclaim: "reproduced Cosmos / matched PAI 0.768 / verified 24/30 / DreamGen VLA win / JEPA child" from smoke alone.

8. Slide Maker brief

Role	Weight / timing guide	Focus
Archaeologist	3 · LONGEST · ~16–18	CLEAR/TENTATIVE; lineage fork Ha/Dreamer/JEPA/LeWM vs Cosmos; family evolution; paradigm foil card; Ablation Bot GAP; exam bite
Visionary	1 · seated Kanta · ~14–15	BEHAVIOR dual-WM; when WFM vs latent; closed-loop/photoreal hooks; hybrid=TENTATIVE; no Discord/Drive
Empiricist	empty seat · ~12–14	H-gate/modes/loop; RC1+RC2+RC4; GPU honesty; must-cite tattoo; exam bite
Stakeholder	empty seat · ~12–14	Buy generative counterweight; risks (44pp, compute, category error); decision ask
Reviewer	empty seat · ~12–14	Tabs.10–13+19–20; verdict Accept systems-WFM; fence closed-loop / DreamGen IF

Spine (paste into Slide Maker) — 5 bullets:

- 1. Generative video WFM as Physical AI world simulator** — Cosmos-Predict2.5 flow T2W/I2W/V2W + Reason1; Transfer2.5 Sim2Real/Real2Real — **PARALLEL** to latent Dreamer/JEPA/LeWM, not their child.
- 2. Must-cite:** PAI **0.768 / 0.810**; Transfer **3.5×**; robot **24/30 vs 5/30**; Bridge PSNR **24.95**; Tab. 20 TimeEmb; DreamGen **91.8** (IF only); AV LET-AP **0.394**; Tab. 9 **4096 H100** MFU — course train **N**.
- 3. Taxonomy fork:** Ha/Dreamer + I-JEPA → V-JEPA → LeWM (**latent plan**) | **Cosmos Predict/Transfer (pixel synth/sim)** | RoboGen (task curriculum).
- 4. Lineage (Archaeologist=3 Nycolas LONGEST):** Predict1 → 2.5 + Transfer1 → 2.5 family evolution; inventum = flow+Reason1+unified modes+apps; always tag latent-plan vs pixel-sim.
- 5. BEHAVIOR'26 + honesty (Visionary=1 Kanta):** Cosmos = photoreal synth/eval appliance; LeWM/JEPA = fast trainable imagination; Empiricist RC1–RC4 infer-only; **do not invent** dual-WM hybrid / DreamGen=VLA-SR as in-paper — **no Discord/Drive**.

GitHub title sketch: [ECE 605] - Cosmos World Simulation Archaeologist Making Three

Full Slide Maker brief: /workspace/world-simulation-quiet/SLIDE_BRIEF.md

Paths

Artifact	Path
This pack	/workspace/world-simulation-quiet/KANTA_PACK.md
PDF	/workspace/world-simulation-quiet/KANTA_PACK.pdf
Slide brief	/workspace/world-simulation-quiet/SLIDE_BRIEF.md
Senior Research	/workspace/world-simulation-quiet/DELIVERABLE.md
Summarizer	/workspace/papers/world-simulation-2511.00062-summary.md
Ablations	AUTHORITATIVE — /workspace/papers/world-simulation-2511.00062-ablations.md (P0: Tab.20 → Tab.17 → Tab.12)
Paper notes	/workspace/world-simulation-quiet/paper_notes.md
Env outlines	/workspace/world-simulation-quiet/env_outline/
Paper PDF	/workspace/papers/world-simulation-2511.00062.pdf
Text extract	/workspace/papers/world-simulation-2511.00062.txt · paper_clean.txt
Page figs	/workspace/papers/cosmos-figs/page-*.png (if present)
Lineage siblings	/workspace/leworldmodel-quiet/ · /workspace/vjepa21-quiet/ · /workspace/ijepa-quiet/ · /workspace/robogen-quiet/
Code	https://github.com/nvidia-cosmos/cosmos-predict2.5 · https://github.com/nvidia-cosmos/cosmos-transfer2.5

Pack synthesized 2026-09-29 PT (HST) for KanBot / ECE 605 — Dec 7 Accompanying Models · Archaeologist=3 (Nykolas) LONGEST · Visionary=1 (Kanta) seated · Empiricist/Stakeholder/Reviewer empty but all five produced · Ablation Bot GAP (paper-extract) · generative video WFM PARALLEL to latent JEPA/LeWM · infer-only course smokes · no Discord/Drive · no invented GPU-hours / DreamGen=VLA-SR / BEHAVIOR hybrid as in-paper.